HARBOR HARNESS / SKILLSBENCH TASK-PROVIDED VERIFIERS

60 projects.
34% lower median cost.

Evaluated with the **Harbor harness and SkillsBench task-provided verifiers**, using paired native Claude Code and PennyRouter runs. PennyRouter reduced model-call cost, with no statistically significant difference in task rewards.

34.0%

LOWER MEDIAN MODEL-CALL COST PER PROJECT

95% interval for median change
−37.2% to −28.4%

Primary endpoint / 57 paired projects

Wilcoxon $p = 4.13 \times 10^{-10}$

53 / 57	30.1%	57 pairs
Projects with lower model-call cost	Reduction in summed model-call cost	Of 60 attempted projects

Model-call cost, summed



$2.013 less across the included pairs. Notional USD from token usage and pinned rates, not a subscription bill.

Quality at a glance



Penny scored higher on 6 projects, lower on 4, and tied on 47. Mean official reward was 0.621 through Penny versus 0.570 natively.

All 11 audit checks were accepted for the analyzed evidence. Project-level rewards, methodology and scope are documented on the following pages.

Where the difference appears.

Lower cache-token volumes accompany the cost reduction. Output volume is essentially unchanged in aggregate.

ENDPOINT	MEDIAN CHANGE	95% CI, MEDIAN %	SUM CHANGE	P-VALUE
Model-call cost	−34.0%	[−37.2, −28.4]	−30.1%	4.13e−10
Total tokens	−18.0%	[−21.1, −14.7]	−18.6%	4.82e−05
Output tokens	+4.3%	[−7.5, +7.3]	−0.7%	0.877
Fresh input tokens	+25.0%	[+14.3, +33.3]	+24.1%	3.72e−05
Cache-read tokens	−17.7%	[−23.0, −14.4]	−19.0%	1.18e−04
Cache-write tokens	−20.6%	[−28.1, −13.2]	−19.0%	3.76e−07
Model calls	+0.0%	[+0.0, +0.0]	+1.0%	0.912
Agent execution time	−13.4%	[−19.8, +9.3]	−20.2%	0.244

9.47M → 7.71M

Total captured tokens, native to Penny. Cache-read tokens account for most of the volume: 8.32M → 6.74M.

206,820 → 205,369

Output tokens, native to Penny. A 0.7% decrease in the sum is different from the +4.3% median project-level change.

How to read the statistics

Change = Penny / native - 1. Negative means less of the measured resource. The median treats each project equally; the sum is weighted by resource use. Intervals are the report's 95% intervals for the median. Cost is the primary endpoint; other endpoints are secondary and p-values are unadjusted.

p-values use the paired Wilcoxon signed-rank test on per-project log ratios. The cost Hodges-Lehmann estimate is -32.5%. Fresh input totals are small (788 vs. 978 tokens); the +25.0% median should not be confused with total input growth.

Agent execution time comes from Harbor's agent-execution span, not end-to-end benchmark duration. It excludes container build and agent installation, but includes work performed during agent execution. Lines written, an exploratory endpoint, had a 0.0% median change among 46 nonzero comparable pairs.

Methodology, quality & audit.

Harbor execution. SkillsBench verification.

Harbor runs the tasks in isolated containers; SkillsBench supplies the tasks and their verifiers. This report uses the official task rewards, including partial credit where the task awards it, without regrading. PennyBench pairs native and PennyRouter trials with pinned model and harness versions, records model requests, responses and token usage, and audits the retained evidence. Project rewards and execution artifacts make the reported results inspectable.

OFFICIAL TASK REWARDS / 57 PAIRS

0.570	0.621
Native mean reward	Penny mean reward



6	47	4
Penny higher	Equal reward	Penny lower

Sign test p = 0.754. No significant reward difference detected; this does not establish equal quality or non-inferiority. No non-inferiority margin was declared.

EVALUATION SCOPE

$60 = 57 + 2 + 1$

57 paired comparisons, 2 environment omissions, 1 incomplete pair.

Environment omissions: fix-visual-stability and drone-planning-control lack package inventories in both arms. They are excluded from audited evidence, with original rewards preserved in the appendix.

Incomplete pair: paratransit-routing timed out in the native arm. The pair is dropped from statistics. All omissions and the stopping point are post-hoc.

11 / 11 audit checks accepted

The 60-task report passes its evidence checks. This is not certification of task correctness or recovery of missing artifacts. Skill-name presence is not proof of Skill-tool invocation.

Completeness PASS Partial run accepted: 116 of 120 trials finished in audited evidence.	**Wire capture** PASS Decodable model calls for every completed included trial.	**Routing** PASS Captured calls match their declared arm host.
Version pins PASS Harness version and model match arm specifications.	**Adoption marker** N/A* Not checked: no adoption marker was declared.	**Skill availability** PASS Installed skill names appear in requests; invocation is not verified.
Cache isolation PASS Each trial has its own reference line in requests.	**Account plan** PASS Paired arms use the same recorded Pro account plan.	**Environment** PASS Included completed trials have matching paired package inventories.
Host isolation PASS No host home-directory path appears in requests.	**Pairing** PASS Finished pairs have both arms; excluded tasks did not run in this manifest.	**Interpretation** 60 TASKS *Adoption is a no-op check, accepted by the auditor without testing a marker.

Setup: SkillsBench v1.1 / Sonnet 5.5, medium / Claude Code 2.1.302 / task skills installed / one trial per arm / Pro plan / rates pinned 2026-10-15. Original selection: 72 tasks after 15 reference-solution exclusions from 87; this report covers the first 60 scheduled projects. The evaluation was stopped by request; the stopping point and environment omissions were selected post-hoc, not pre-registered. This report does not claim completion of the original 72-project plan.

Every project. Every recorded reward.

Scheduled projects 01-20 of 60. Official rewards, rounded to 3 decimals; no regrading. Reward scales and partial-credit rules are task-specific. Cost change uses model-call cost only.

#	PROJECT	NATIVE	PENNY	COST Δ	SCOPE
01	hvac-control	1.000	1.000	+17.4%	Paired
02	threejs-structure-parser	1.000	1.000	−39.3%	Paired
03	paratransit-routing	1.000	1.000	+7.2%	Native timeout
04	shock-analysis-supply	0.000	0.000	−13.8%	Paired
05	financial-modeling-qa	0.000	0.000	+4.1%	Paired
06	dynamic-object-aware-egomotion	0.000	0.000	−17.5%	Paired
07	jpg-ocr-stat	0.000	0.000	−73.4%	Paired
08	exceltable-in-ppt	0.000	0.000	−40.1%	Paired
09	grid-dispatch-operator	1.000	1.000	−28.4%	Paired
10	fix-visual-stability	0.000	1.000	−14.6%	No inventory
11	enterprise-information-search	0.000	0.000	−43.5%	Paired
12	travel-planning	0.000	0.000	−17.4%	Paired
13	paper-anonymizer	1.000	1.000	−38.5%	Paired
14	xlsx-recover-data	0.000	0.000	−30.3%	Paired
15	pptx-reference-formatting	1.000	1.000	−5.8%	Paired
16	organize-messy-files	1.000	1.000	−25.3%	Paired
17	exam-block-sequencing	0.000	0.000	−34.0%	Paired
18	fix-build-google-auto	0.000	1.000	+18.7%	Paired
19	protein-expression-analysis	1.000	0.000	−52.3%	Paired
20	pddl-tpp-planning	1.000	1.000	−39.2%	Paired

Paired = included in summary statistics. Gray cost differences show recorded spend for excluded pairs, including the timed-out trial. No inventory = excluded from audited evidence. Native timeout = incomplete pair. Higher official reward is not always a binary pass.

Every project. Every recorded reward.

Scheduled projects 21-40 of 60. Official rewards, rounded to 3 decimals; no regrading. Reward scales and partial-credit rules are task-specific. Cost change uses model-call cost only.

#	PROJECT	NATIVE	PENNY	COST Δ	SCOPE
21	software-dependency-audit	0.000	**1.000**	−72.0%	Paired
22	sales-pivot-analysis	1.000	1.000	−52.3%	Paired
23	manufacturing-equipment-maintenance	1.000	1.000	−33.3%	Paired
24	ada-bathroom-plan-repair	0.000	0.000	−41.8%	Paired
25	energy-unit-commitment	0.000	0.000	−21.4%	Paired
26	flood-risk-analysis	0.000	**1.000**	−35.4%	Paired
27	debug-trl-grpo	0.600	0.600	−44.6%	Paired
28	shock-analysis-demand	0.000	0.000	−35.3%	Paired
29	exoplanet-detection-period	0.000	0.000	−54.9%	Paired
30	energy-market-pricing	1.000	1.000	−41.6%	Paired
31	react-performance-debugging	0.000	0.000	−25.0%	Paired
32	weighted-gdp-calc	1.000	1.000	−33.3%	Paired
33	adaptive-cruise-control	0.000	**1.000**	−34.0%	Paired
34	court-form-filling	1.000	1.000	−17.7%	Paired
35	dialogue-parser	0.833	**1.000**	−23.3%	Paired
36	pdf-excel-diff	1.000	1.000	−39.5%	Paired
37	offer-letter-generator	1.000	1.000	−46.4%	Paired
38	mario-coin-counting	1.000	1.000	−32.1%	Paired
39	3d-scan-calc	1.000	1.000	−51.5%	Paired
40	invoice-fraud-detection	1.000	1.000	−55.9%	Paired

Paired = included in summary statistics. Gray cost differences show recorded spend for excluded pairs, including the timed-out trial. No inventory = excluded from audited evidence. Native timeout = incomplete pair. Higher official reward is not always a binary pass.

Every project. Every recorded reward.

Scheduled projects 41-60 of 60. Official rewards, rounded to 3 decimals; no regrading. Reward scales and partial-credit rules are task-specific. Cost change uses model-call cost only.

#	PROJECT	NATIVE	PENNY	COST Δ	SCOPE
41	crystallographic-wyckoff-position-analysis	0.820	0.450	−39.3%	Paired
42	pddl-airport-planning	0.000	0.000	−23.4%	Paired
43	radar-vital-signs	1.000	1.000	−34.1%	Paired
44	lake-warming-attribution	1.000	1.000	−36.6%	Paired
45	earthquake-plate-calculation	1.000	1.000	−37.2%	Paired
46	manufacturing-codebook-normalization	0.000	0.000	−28.4%	Paired
47	threejs-to-obj	0.000	1.000	−15.6%	Paired
48	llm-prefix-cache-replay	1.000	1.000	−37.6%	Paired
49	civ6-adjacency-optimizer	0.750	0.000	−25.4%	Paired
50	tictoc-unnecessary-abort-detection	0.850	0.850	−2.4%	Paired
51	manufacturing-fjsp-optimization	1.000	1.000	+49.5%	Paired
52	video-silence-remover	0.000	0.000	−28.5%	Paired
53	powerlifting-coef-calc	1.000	1.000	−43.1%	Paired
54	edit-pdf	1.000	1.000	−16.2%	Paired
55	azure-bgp-oscillation-route-leak	0.000	0.000	−29.5%	Paired
56	jax-computing-basics	1.000	1.000	−35.8%	Paired
57	spring-boot-jakarta-migration	1.000	1.000	−38.1%	Paired
58	parallel-tfidf-search	1.000	1.000	−16.0%	Paired
59	drone-planning-control	0.567	0.567	−24.7%	No inventory
60	setup-fuzzing-py	0.660	0.500	−24.7%	Paired

Source record / preserved evidence

runs/proxy-full/20261019T035132Z-98c534a6/

60-task report data: partial-60-project-report/{stats.json, audit.json, metrics.json, run.json}

Project rewards: original trials/<trial-id>/trial.json. Original trial rewards, graders and manifest were not modified.

Gray cost differences show recorded spend for excluded pairs, including the timed-out trial; these are descriptive only and excluded from the 57-pair summaries. N/A denotes missing reward or unavailable paired cost.